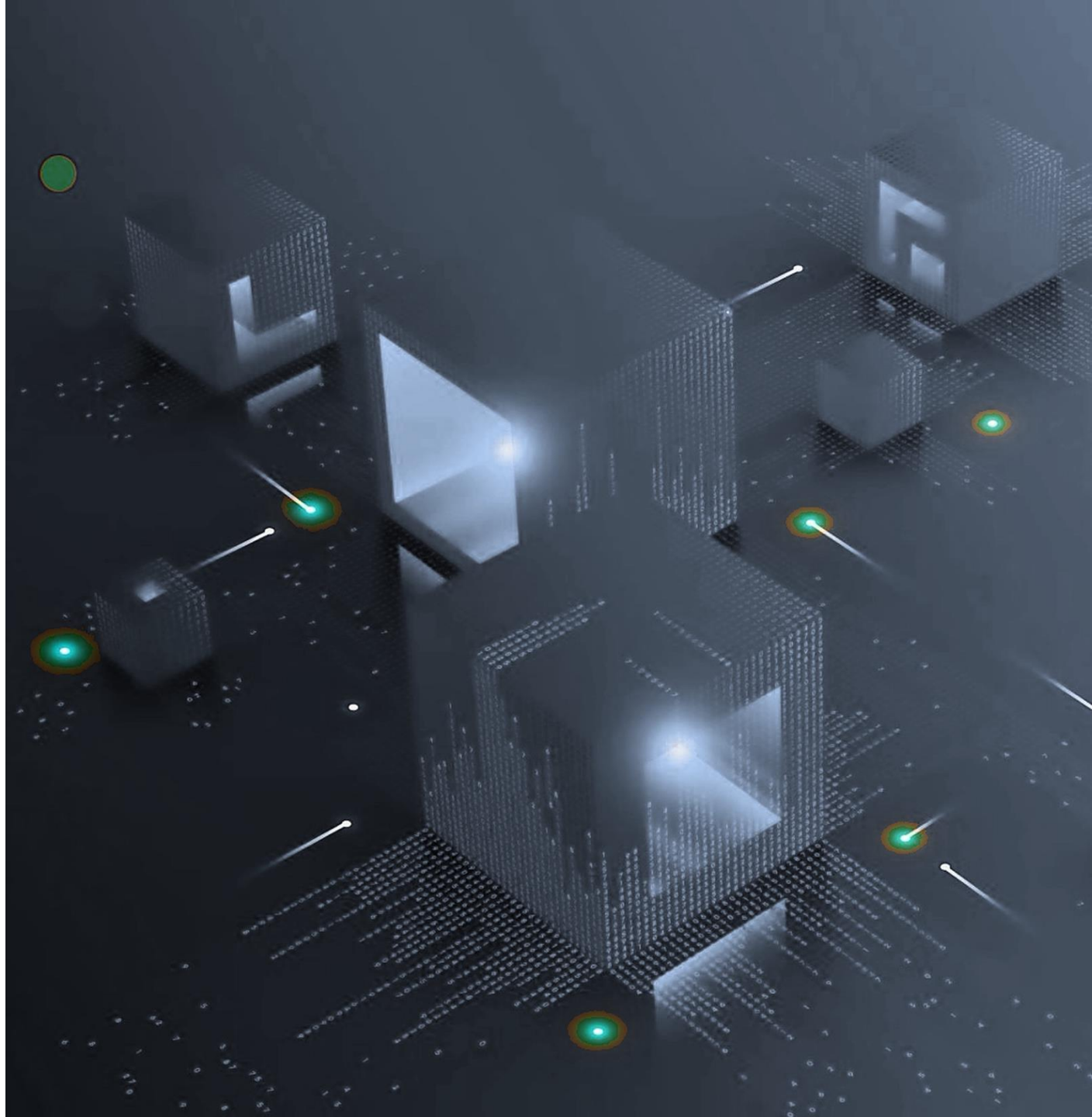
ScholarPlan: An LLM-Powered Planning Agent for Academic Research and Information Gathering

MSc Artificial Intelligence

Module: Intelligent Agents

University of Essex



PROBLEM & OBJECTIVES

THE PROBLEM

- Finding relevant academic sources
- Managing large amounts of information
- Turning evidence into structured findings
- Verifying generated claims
- Repeating searches when evidence is insufficient

THE MOTIVATION

- Structured planning
- Evidence-grounded retrieval
- Independent verification
- Adaptive re-planning
- Traceable execution

STANDARD LLM

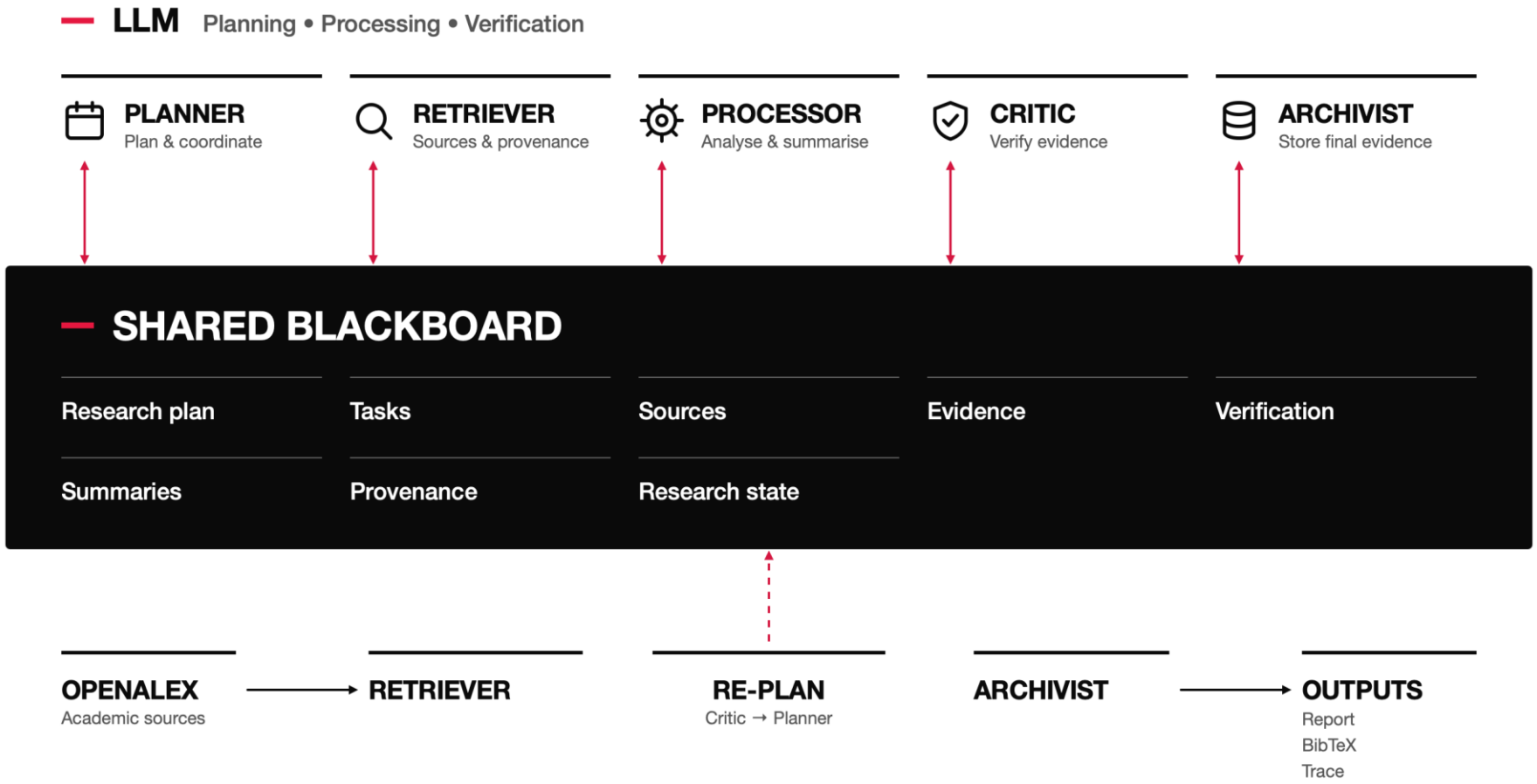


One shot: no plan, no checks, no trace.

SCHOLARPLAN



SYSTEM ARCHITECTURE



HOW THE AGENT WORKS



01. Research Goal

Investigate a research question



02. Plan & Act

Retrieve sources and generate claims



03. Evaluate Evidence

Critic checks support and relevance

Evidence insufficient	Stopping criteria met
Feedback -> revised plan -> retrieve again	Archive report, references and trace

IMPLEMENTATION

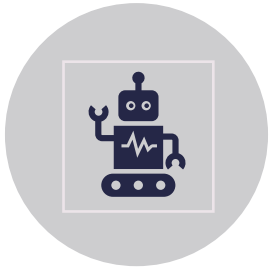


LLM ORCHESTRATION

PYTHON

AZURE AI

GPT-5-MINI



RETRIEVAL

OPENALEX API

ACADEMIC SOURCES



STATE & MEMORY

SQLITE BLACKBOARD

RUN ISOLATION



SEMANTIC ANALYSIS

ALL-MINILM-L6-V2

384D EMBEDDINGS

NUMPY



Outputs

Research Report · BibTeX · Execution Trace · Evaluation Metrics

DEMONSTRATION: SUCCESSFUL RUN

Research goal:
Investigate the impact of LLMs on academic research.

40
Sources retrieved

17
Claims supported

19
Claims generated

2
Iterations completed

✓ 89.5% Critic-supported claim rate

17 of 19 claims classified as supported

```
============================================================
SCHOLARPLAN FINAL STATUS
============================================================
Status: completed
Iterations executed: 2
Sources retrieved: 40
Claims generated: 19



============================================================
SCHOLARPLAN EVALUATION
============================================================
Claims evaluated: 19
Supported claims: 17
Unsupported claims: 0
Irrelevant claims: 2
Insufficient evidence: 0
Support rate: 89.5%
```

AUTONOMOUS RE-PLANNING

EXECUTION TRACE

```
Event 539  REPLANNING_STARTED
        feedback_count: 10

Event 540  PLAN_CREATED
        iteration: 2
        based_on_feedback: true
```



Step 1 – Evidence Evaluation

Checks evidence support and research relevance



Step 2 – Feedback

Verification feedback informs the next plan



Step 3 – Re-Planning

Planner creates revised tasks and queries



Step 4 – Second Iteration

Retrieve -> Process -> Verify

TESTING & EVALUATION

24

Tests passed

Unit and functional tests

89.5%

Supported claim rate

17 of 19 claims supported

100%

Embedding coverage

40 of 40 retrieved sources

2

Iterations completed

Feedback-driven re-planning

Testing approach

Unit tests | Functional tests | End-to-end execution

Evaluation checks pipeline behaviour, evidence support and output generation.

CRITICAL EVALUATION

24 tests passed

STRENGTHS

- ✓ Autonomous planning and re-planning
- ✓ Shared SQLite Blackboard
- ✓ Traceable execution and saved outputs

LIMITATIONS

- ⚠ Critic verification is not ground truth
- ⚠ Retrieval depends on OpenAlex
- ⚠ Maximum of 3 iterations
- ⚠ No independent human evaluation

FUTURE IMPROVEMENTS

1. Integrate additional academic databases
2. Benchmark claim verification against human assessments
3. Evaluate retrieval quality and duplicate detection more rigorously

CONCLUSION + REFERENCES

SCHOLARPLAN · AUTONOMOUS ACADEMIC RESEARCH

Autonomous workflow

Planning, retrieval, claim processing, verification and re-planning.

Evidence-grounded outputs

Academic sources, structured claims, references and execution traces.

Next step

Stronger independent verification and broader academic retrieval.

KEY REFERENCES

Yao, S. et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models.

Shinn, N. et al. (2023). Reflexion: Language Agents with Verbal Reinforcement Learning.

Reimers, N. and Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.

Full references and technical documentation are available in the project README.